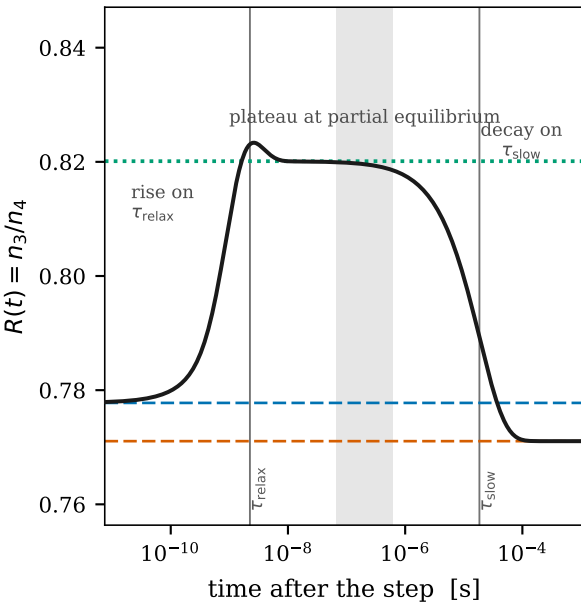
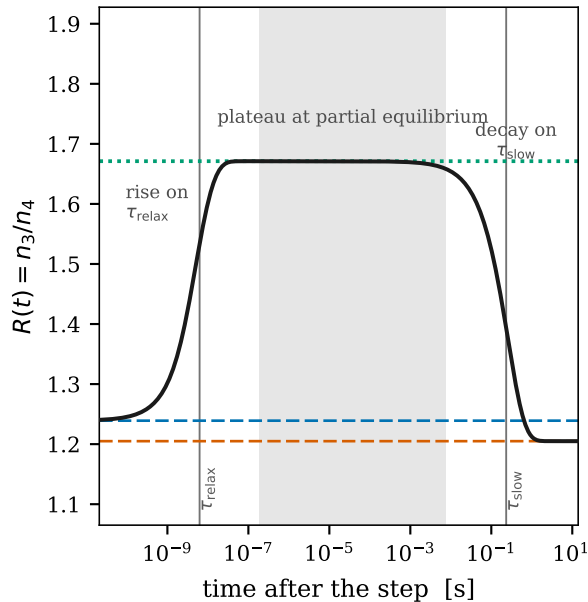


(a) benchmark:  $T_e$  2.95  $\rightarrow$  3.09 eV,  $n_e = 1.4 \times 10^{14} \text{ cm}^{-3}$



(b) coldest grid point:  $T_e$  1.00  $\rightarrow$  1.05 eV,  $n_e = 5.2 \times 10^{13} \text{ cm}^{-3}$



- $R(t)$ , 43-state solution
- ⋯  $R^{\text{PE}}$ , partial equilibrium (one linear solve)
- -  $R^{\text{CRE}}$  at  $T_e$  before the step
- -  $R^{\text{CRE}}$  at  $T_e$  after the step
- plateau window,  $30\tau_{\text{relax}}$  to  $\tau_{\text{slow}}/30$